DATA LITERACY 101 - MUST KNOW CONCEPTS

Comprehensive Study Guide for Core Data Engineering Manager Role (FAANG Preparation)

🧠 1. Core Data Concepts

1. **Data**: Raw facts, figures, or symbols (e.g., numbers, text, images) without context. It is the foundational input.
2. **Information**: Processed, organized, or contextualized data that provides meaning (e.g., sales figures grouped by region become regional performance insights).
3. **Knowledge**: Applied information with understanding, experience, or insight for decision-making (e.g., knowing why sales dropped in one region and how to act).
4. **Metadata**: "Data about data" — describes structure, content, context, or usage (e.g., file creation date, column data type, source system). Types include technical, business, and operational.
5. **Master Data**: Core, shared business entities that are relatively stable (e.g., customers, products, employees). Provides a consistent view across systems.
6. **Reference Data**: Standardized codes or lookup values (e.g., country codes, currency codes, product categories). Often used to classify master or transaction data.
7. **Transaction Data**: Records of business events or activities (e.g., orders, payments, clicks). High volume, time-sensitive, and changes frequently.
8. **Operational Data**: Data used for day-to-day business operations (often from OLTP systems). Supports transactional processing.
9. **Analytical Data**: Data prepared for analysis, reporting, and decision support (often aggregated or historical from data warehouses/lakes).
10. **Raw Data**: Unprocessed data in its original form (e.g., logs from sensors or raw API responses). Flexible but requires heavy transformation.
11. **Curated Data**: Cleaned, enriched, and validated data ready for consumption (e.g., after ETL/ELT and quality checks).
12. **Trusted Data**: High-quality, governed, and reliable data that stakeholders can confidently use (often tied to "golden records").
13. **Golden Record**: The single, authoritative, best version of a master data entity (e.g., the most accurate customer profile after resolving duplicates and conflicts). Achieved via master data management.
14. **Event Data**: Discrete occurrences in event-driven systems (common in streaming).

Interview Tip: Always discuss the **Data** → **Information** → **Knowledge** progression. Master/Reference data adds context to Transaction data. In manager interviews, link to business value (e.g., Customer 360 as golden record).

🏛️ 2. Data Governance Terms

1. **Data Governance**: Framework of policies, processes, roles, and standards to manage data as a strategic asset. Ensures quality, security, compliance, and value. Central to DAMA-DMBOK.
2. **Data Ownership**: Accountability for data quality, accuracy, and appropriate use. Business owners (not IT) typically own data.
3. **Data Stewardship**: Operational role focused on day-to-day data quality, definitions, and issue resolution. Stewards implement policies.
4. **Data Custodian**: Technical role responsible for safe storage, backup, and access management of data (more IT-focused than stewardship).
5. **Data Accountability**: Clear assignment of responsibility for data outcomes (often via RACI matrices).
6. **Data Policy**: High-level rules guiding data handling (e.g., "all sensitive data must be encrypted").
7. **Data Standard**: Specific, enforceable requirements (e.g., "customer names must follow title case and be stored in UTF-8").
8. **Data Compliance**: Adherence to laws, regulations, and internal policies (e.g., data retention rules).
9. **Data Ethics**: Moral considerations in data use (e.g., fairness, avoiding bias, transparency in AI decisions).
10. **Data Privacy**: Protection of personal/sensitive information from unauthorized access or misuse.
11. **Data Access Control**: Mechanisms to restrict who can view or modify data (e.g., RBAC, ABAC).
12. **Data Classification**: Categorizing data by sensitivity (e.g., public, internal, confidential, restricted) to apply appropriate controls.
13. **Data Retention Policy**: Rules defining how long data must be kept and when it should be deleted/archived.
14. **Data Governance Council**: Cross-functional body (executives, stewards, owners) that oversees governance strategy and resolves issues.

Interview Tip: Governance enables trust and scalability at FAANG scale. As a manager, emphasize operating models (centralized vs. federated vs. Data Mesh), RACI in practice, and how you've driven adoption across domains. Include regulations such as GDPR, HIPAA, and SOC2.

🧪 3. Data Quality Terms

Data Quality: Fitness for purpose—data is accurate, complete, etc., for its intended use. Measured via dimensions.

Key Data Quality Dimensions (6 core ones):

1. **Accuracy**: Data correctly represents real-world facts (e.g., correct customer address).
2. **Completeness**: No missing values where required (e.g., all orders have a customer ID).

3. **Consistency**: Data is uniform across systems/sources (e.g., same customer name format everywhere).
4. **Timeliness**: Data is up-to-date and available when needed (e.g., real-time sales data).
5. **Validity**: Data conforms to defined rules/formats (e.g., email follows standard pattern).
6. **Uniqueness**: No unintended duplicates (e.g., one record per customer).

Additional Important Dimensions:

1. **Data Integrity**: Overall reliability, including referential integrity (no orphan records).
2. **Data Reconciliation**: Process of verifying data consistency between systems (e.g., source vs. target totals match).

Related Processes:

1. **Data Profiling**: Analyzing data to understand structure, content, quality issues (statistics, patterns, anomalies).
2. **Data Cleansing (Cleaning)**: Fixing errors, filling gaps, correcting values.
3. **Data Validation**: Checking data against rules during ingestion/transformation.
4. **Data Enrichment**: Adding external or derived information (e.g., appending geolocation to addresses).
5. **Data Standardization**: Enforcing consistent formats (e.g., dates as YYYY-MM-DD).
6. **Data Deduplication**: Identifying and merging/removing duplicates.

Interview Tip: Discuss measurement (profiling tools, observability), remediation (quality gates in pipelines), and manager-level accountability (SLAs, alerting, RCA for quality incidents).

🏗️ 4. Data Architecture Terms

1. **Data Architecture**: Blueprint for how data is collected, stored, processed, and consumed across an organization. Aligns with business needs.
2. **Data Model**: Abstract representation of data entities, attributes, and relationships.
    - **Conceptual Data Model**: High-level, business-oriented (entities and relationships, no technical details).
    - **Logical Data Model**: Detailed, independent of technology (attributes, keys, normalization).
    - **Physical Data Model**: Implementation-specific (tables, columns, data types, indexes, partitions).
3. **Data Schema**: Structure defining tables, columns, constraints, and relationships in a database.
4. **Data Warehouse Schema**:
    - **Star Schema**: Central fact table (measures) connected to denormalized dimension tables. Simple, fast queries, higher storage due to redundancy.
    - **Snowflake Schema**: Normalized version of star — dimension tables split into sub-tables. Saves storage but requires more joins (slower queries).

5. **Data Mart**: Subset of data warehouse focused on a specific business area (e.g., marketing or sales mart). Can be dependent (from warehouse) or independent.
6. **Data Lake**: Centralized repository for raw data in native format (structured, semi-structured, unstructured). Schema-on-read, cheap storage, suited for big data/ML.
7. **Data Lakehouse**: Hybrid of data lake (flexible, raw storage) and data warehouse (ACID transactions, schema enforcement, BI support). Adds reliability to lakes (e.g., via Delta Lake, Iceberg).
8. **Data Mesh**: Decentralized, domain-oriented architecture where domains treat data as products. Focuses on organizational change and federated governance.
9. **Data Fabric**: Technology-driven, centralized approach using automation and metadata to unify data access/integration across hybrid environments.
10. **Entity Relationship (ER) Model**: Visual diagram showing entities, attributes, and relationships (used in conceptual/logical modeling).

Comparison Tip: Star (speed) vs Snowflake (storage efficiency). Data Lake (raw/flexible) vs Warehouse (structured/optimized) vs Lakehouse (best of both). Data Mesh (decentralized, domain-owned) vs Data Fabric (centralized, automated).

2026 Tip: Lakehouse (with Iceberg as ecosystem winner) is dominant. Discuss separation of storage and compute as a core cloud advantage.

🔄 5. Data Engineering Terms

1. **ETL (Extract, Transform, Load)**: Traditional pattern — transform data before loading into target (e.g., warehouse). Good for structured data, but can be resource-heavy.
2. **ELT (Extract, Load, Transform)**: Modern pattern (common in cloud) — load raw data first, then transform in the target (leverages scalable compute). Better for big data/lakes. **Trade-offs**: latency, scalability, cost.
3. **Data Pipeline**: Automated workflow moving and transforming data from source to destination. Includes ingestion, processing, loading.
4. **Data Ingestion**: Process of bringing data into a system (batch or streaming).
5. **Batch Processing**: Handling data in groups at scheduled intervals (e.g., nightly jobs). Efficient for large volumes.
6. **Real-time Processing**: Immediate or near-immediate data handling.
7. **Stream Processing**: Continuous processing of data in motion (e.g., events from sensors).
8. **Data Orchestration**: Coordinating and scheduling complex pipelines (tools like Airflow, Dagster, Prefect, cloud-native like Step Functions, Composer, Data Factory).
9. **Data Transformation**: Converting data format/structure (cleaning, aggregating, joining).
10. **Data Integration**: Combining data from disparate sources into a unified view.
11. **Data Replication**: Copying data from one system to another (for redundancy or access).
12. **Data Sync**: Keeping data consistent across systems (often bidirectional).
13. **Data Migration**: Moving data between systems/storage (e.g., on-prem to cloud).
14. **Data Pipeline Monitoring**: Tracking performance, errors, data volumes, and SLAs.
15. **Python for DE**: PySpark, pandas optimization, concurrency, testing with pytest.

16. **Containerization**: Docker + Kubernetes for Spark/Flink jobs.
17. **Monitoring and Alerting**.

Interview Tip: Know ETL/ELT deeply. As a manager, discuss orchestration reliability, backpressure handling, and choosing tools based on team skills and scale.

🧾 6. Metadata & Catalog Terms

1. **Metadata Management**: Processes to collect, store, and use metadata for better data understanding.
2. **Data Catalog**: Searchable inventory of data assets with metadata, lineage, and business context. Enables discovery.
3. **Business Glossary**: Centralized definitions of business terms (e.g., what "Active Customer" means).
4. **Technical Metadata**: System-oriented (schemas, data types, pipelines).
5. **Business Metadata**: User-oriented (definitions, owners, usage rules).
6. **Operational Metadata**: Runtime info (e.g., job run times, errors).
7. **Data Dictionary**: Detailed documentation of data elements (names, types, descriptions, constraints).
8. **Data Lineage**: Tracking data flow from origin to consumption (shows transformations and dependencies).
9. **Data Traceability**: Ability to trace data back to its source (related to lineage).
10. **Data Discovery**: Finding relevant data assets using catalogs/search.
11. **Data Tagging**: Labeling assets with keywords or categories for organization/search.
12. **Data Asset Inventory**: Comprehensive list of all data assets with ownership and metadata.

Interview Tip: Lineage is critical for impact analysis, compliance, and troubleshooting at scale. Mention tools like DataHub, Amundsen, Unity Catalog, or Snowflake's native capabilities.

🔐 7. Data Security & Privacy Terms

1. **Data Security**: Protecting data from unauthorized access, corruption, or loss.
2. **Data Encryption**: Converting data to unreadable format (at-rest or in-transit).
3. **Data Masking**: Replacing sensitive data with realistic but fake values for non-production use.
4. **Tokenization**: Replacing sensitive data with non-sensitive tokens (reversible with vault).
5. **Anonymization**: Irreversibly removing personal identifiers so individuals cannot be re-identified.
6. **Pseudonymization**: Replacing identifiers with pseudonyms (reversible with additional info; GDPR-friendly).
7. **Role-Based Access Control (RBAC)**: Access based on user roles (e.g., analyst can read but not delete).
8. **Attribute-Based Access Control (ABAC)**: Dynamic access based on attributes (user, data, context).

9. **Data Loss Prevention (DLP)**: Tools/policies to detect and prevent sensitive data leakage.
10. **Audit Trail**: Log of all data access/modification for compliance and forensics.
11. **Data Breach**: Unauthorized access or exposure of data.
12. **GDPR**: EU regulation on data protection and privacy (rights like erasure, consent).
13. **HIPAA Compliance**: US rules for protecting health information.
14. **SOC2 Compliance**: Controls for security, availability, processing integrity, confidentiality, and privacy (common for SaaS).

Interview Tip: Differentiate privacy techniques (Anonymization vs. Pseudonymization). At manager level: How you embed security in architecture and governance (data classification driving controls).

📊 8. Data Analytics Terms

1. **Data Analytics**: Process of examining data to draw conclusions and support decisions.
2. **Business Intelligence (BI)**: Tools and practices for transforming data into actionable insights (reports, dashboards).
3. **Descriptive Analytics**: What happened? (summarizes past data).
4. **Diagnostic Analytics**: Why did it happen? (root cause via drilling down).
5. **Predictive Analytics**: What will happen? (forecasting with models).
6. **Prescriptive Analytics**: What should we do? (recommendations with optimization).
7. **Data Visualization**: Graphical representation of data (charts, graphs).
8. **Dashboard**: Interactive visual summary of key metrics for monitoring.
9. **KPI (Key Performance Indicator)**: Measurable value showing progress toward business goals.
10. **Metrics**: Quantifiable measures (KPI is a type of metric).
11. **Dimension**: Descriptive attribute for slicing data (e.g., time, region, product).
12. **Fact Table**: Central table in star/snowflake schema containing measurable, numeric data (e.g., sales amount).
13. **OLAP (Online Analytical Processing)**: Multidimensional analysis on historical data (cubes, slicing, dicing). Optimized for complex queries.
14. **OLTP (Online Transaction Processing)**: High-volume, fast transactional operations (inserts/updates). Normalized schemas.

Interview Tip: OLTP (normalized, fast transactions) vs. OLAP (denormalized, complex queries). Link to how DE enables self-service analytics.

🧮 9. Data Modeling Terms

1. **Data Model**: See section 4.
2. **Entity**: Real-world object or concept (e.g., Customer).
3. **Attribute**: Property of an entity (e.g., CustomerName, Email).
4. **Relationship**: Association between entities (e.g., Customer places Order).
5. **Primary Key**: Unique identifier for a record in a table.

6. **Foreign Key**: Field referencing primary key in another table (enforces relationships).
7. **Composite Key**: Primary key made of multiple columns.
8. **Surrogate Key**: System-generated unique ID (e.g., auto-increment integer) instead of natural/business key.
9. **Cardinality**: Number of relationships (one-to-one, one-to-many, many-to-many).
10. **Normalization**: Organizing data to reduce redundancy and improve integrity (1NF, 2NF, 3NF, BCNF). Removes anomalies.
11. **Denormalization**: Intentionally adding redundancy for query performance (common in warehouses).
12. **Normal Forms**:
    - **1NF**: Atomic values, no repeating groups.
    - **2NF**: 1NF + no partial dependency on composite keys.
    - **3NF**: 2NF + no transitive dependencies.
    - **BCNF**: Stronger 3NF (every determinant is a candidate key).
13. **Constraint**: Rules enforced by database (NOT NULL, UNIQUE, CHECK, FOREIGN KEY).
14. **Indexing**: Data structure to speed up queries (e.g., B-tree on columns frequently filtered).

Interview Tip: Normalization for OLTP integrity; Denormalization for OLAP speed. SCD Types (especially Type 2) are frequently tested. Be ready to model a business scenario on the spot.

☁️ 10. Cloud & Modern Data Platforms

1. **Cloud Data Platform**: Managed services for storage, processing, and analytics in the cloud (e.g., AWS, Azure, GCP offerings).
2. **Data Lake / Data Warehouse / Data Lakehouse**: See section 4.
3. **Snowflake Architecture**: Cloud data platform with separation of storage and compute (elastic scaling). Supports multi-cluster warehouses.
4. **Multi-cloud**: Using services from multiple cloud providers.
5. **Hybrid Cloud**: Combination of on-premises and cloud infrastructure.
6. **Serverless Data Processing**: Pay-per-use compute without managing servers (e.g., AWS Lambda, Snowflake serverless).
7. **Data Platform-as-a-Service (DPaaS)**: Fully managed platform for data needs.
8. **Big Data**: Extremely large, complex datasets requiring specialized tools for storage/processing (volume, velocity, variety, veracity — the 4Vs).

Key Services:

- **AWS**: S3, Glue, EMR, Redshift, Kinesis, MSK, Lake Formation, Step Functions.
- **GCP**: BigQuery, Dataflow, Pub/Sub, Composer, Datastream.
- **Azure**: Data Factory, Synapse, Event Hubs, ADLS + Delta Lake.
- **Cross-cloud**: Databricks (Unity Catalog, Delta Live Tables), Snowflake (multi-cluster warehouses, separation of storage/compute).

Modern Data Stack (MDS): ELT-focused (Fivetran/Airbyte + dbt + Snowflake/Databricks/BigQuery + Looker/Tableau). 2026 trend: consolidation and lakehouse-first.

Cost Optimization / FinOps: Partitioning, clustering, auto-scaling, spot instances, storage tiering.

Interview Tip: Always discuss trade-offs (e.g., Snowflake vs. Databricks vs. BigQuery) and separation of storage and compute for elastic scaling.

⚡ 11. Big Data Technologies

1. **Hadoop**: Framework for distributed storage and processing of big data.
2. **HDFS (Hadoop Distributed File System)**: Fault-tolerant, scalable storage across clusters.
3. **MapReduce**: Programming model for parallel processing of large datasets.
4. **Spark**: In-memory processing engine (faster than MapReduce). Supports batch, streaming, ML, SQL, and Delta integration.
5. **Hive**: SQL-like querying on Hadoop (data warehousing on HDFS).
6. **Kafka**: Distributed streaming platform for real-time data pipelines (publish-subscribe).
7. **Flink**: Stream-processing framework with strong support for event-time processing and state management.
8. **NoSQL**: Non-relational databases for flexible schemas and scalability (vs traditional RDBMS).
9. **MongoDB**: Document-oriented NoSQL (JSON-like documents).
10. **Cassandra**: Wide-column store for high write throughput and availability.
11. **ElasticSearch**: Search and analytics engine (often for logs, full-text search).

Added: Lakehouse table formats (**Delta Lake, Apache Iceberg, Hudi**) — provide ACID, time travel, schema evolution.

Interview Tip: Spark is versatile (batch + streaming); Kafka for ingestion/streaming. Discuss skew mitigation, dynamic allocation, and containerized jobs.

🤖 12. AI / ML Data Terms

1. **Training Data**: Data used to train ML models (largest portion).
2. **Test Data**: Held-out data to evaluate final model performance.
3. **Validation Data**: Used during training to tune hyperparameters and prevent overfitting.
4. **Feature Engineering**: Creating/transforming input variables (features) to improve model performance.
5. **Feature Store**: Centralized repository for storing, managing, and serving features for ML (online/offline serving, point-in-time correctness).
6. **Labeling**: Assigning target/outcome values to data for supervised learning.
7. **Model Training**: Process of fitting a model to training data.

8. **Model Drift**: Degradation in model performance over time due to changes in data distribution.
9. **Data Bias**: Systematic errors in data leading to unfair or skewed model outcomes.
10. **Data Sampling**: Selecting a subset of data (random, stratified) for efficiency or balance.
11. **Supervised Learning**: Training with labeled data (e.g., regression, classification).
12. **Unsupervised Learning**: Finding patterns in unlabeled data (e.g., clustering, dimensionality reduction).

Added: Vector databases (for embeddings), **RAG data pipelines** (chunking, freshness, embedding stores), Model data lineage & monitoring.

Interview Tip: DE's role in production ML includes reliable feature pipelines and drift detection.

🧭 13. Data Lifecycle Terms

1. **Data Creation/Acquisition**: Generating or collecting data.
2. **Data Storage**: Persisting data in suitable systems.
3. **Data Processing**: Transforming and preparing data.
4. **Data Usage**: Analysis, reporting, ML, etc.
5. **Data Archival**: Moving inactive data to cheaper, long-term storage.
6. **Data Retention**: Keeping data for required periods (compliance-driven).
7. **Data Deletion**: Securely removing data when no longer needed.
8. **Data Archiving Policy**: Rules for when and how to archive/delete data.

Interview Tip: Lifecycle management ties directly to retention policies, compliance, and cost optimization.

🏦 14. Enterprise & Domain Data Terms

1. **Customer Data**: Information about individuals/organizations (often sensitive — privacy focus).
2. **Product Data**: Details about goods/services (master data).
3. **Policy Data (Insurance)**: Contract/terms information.
4. **Claims Data (Insurance)**: Records of incidents and payouts (transactional).
5. **Financial Data**: Accounting, transactions, ledgers.
6. **Transaction History**: Chronological record of events.
7. **Ledger Data**: Immutable record of financial transactions (e.g., blockchain-inspired in modern systems).
8. **Account Master**: Core account information (master data).
9. **Customer 360**: Unified, comprehensive view of a customer across all touchpoints (requires master data management).
10. **Single Source of Truth (SSOT)**: Authoritative, consistent data source for a domain (often achieved via golden records).

🔁 15. Data Operations Terms (DataOps)

1. **DataOps**: Applying DevOps principles to data — automation, collaboration, and agility for data pipelines.
2. **CI/CD for Data**: Continuous Integration/Continuous Deployment pipelines for data (testing schemas, transformations).
3. **Data Pipeline Automation**: Scheduling and orchestrating pipelines without manual intervention.
4. **Data Observability**: Monitoring data for quality, freshness, volume, and schema changes (like application observability).
5. **Data Monitoring**: Tracking pipeline health and data metrics.
6. **Data Alerts**: Notifications for anomalies or SLA breaches.
7. **Data SLAs**: Service Level Agreements defining expected data quality, timeliness, etc.
8. **Data Reliability**: Ensuring pipelines deliver trustworthy data consistently.
9. **Incident Management**: Handling data-related outages or quality issues.
10. **Root Cause Analysis (RCA)**: Systematic investigation of incidents to prevent recurrence.

Interview Tip: DataOps emphasizes reliability engineering — treat pipelines like software. Manager focus: designing on-call, SLAs, and observability strategy.

🧾 16. Governance Framework Terms

1. **DAMA-DMBOK**: Data Management Body of Knowledge from DAMA International. Comprehensive guide with 11 knowledge areas (Data Governance at center): Governance, Architecture, Modeling & Design, Storage & Operations, Security, Integration & Interoperability, Documents & Content, Reference & Master Data, Warehousing & BI, Metadata, Data Quality. Includes ethics, maturity models, and best practices. Widely used for CDMP certification.
2. **CDO (Chief Data Officer)**: Executive responsible for data strategy, governance, and monetization.
3. **Data Governance Operating Model**: Defines how governance is executed (centralized, decentralized, federated).
4. **RACI Matrix**: Tool assigning Responsible, Accountable, Consulted, Informed roles for data activities.
5. **Data Strategy**: Roadmap aligning data initiatives with business goals.
6. **Data Maturity Model**: Assesses organization's data capabilities (e.g., initial, managed, optimized levels).
7. **Data Ownership Model**: Framework defining who owns what data.
8. **Data Stewardship Model**: Structure for stewardship roles (central vs embedded).

Added Depth on Data Mesh: Domain-owned data products, federated computational governance, data contracts, and SLAs.

ADVANCED & MANAGER-LEVEL CONCEPTS

🔄 17. Advanced Data Engineering & Pipeline Concepts (Most Critical)

1. **Change Data Capture (CDC)**: Techniques (e.g., log-based, trigger-based) to detect and propagate only incremental changes (inserts/updates/deletes) from source systems. Essential for efficient incremental loads instead of full refreshes.
2. **Idempotency**: Property where re-running the same operation produces the same result without side effects (duplicates). Critical for reliable pipelines.
3. **Exactly-Once Semantics**: Delivery guarantee in streaming systems (vs at-least-once or at-most-once). Discuss implementation challenges with Kafka + Spark/Flink.
4. **Backfill**: Reprocessing historical data for a pipeline (e.g., after bug fix or new logic). Often paired with idempotency and partitioning.
5. **Slowly Changing Dimensions (SCD)**: Handling changes in dimension tables over time (Type 1: overwrite; Type 2: add new row with version; Type 3: add column, etc.).
6. **Schema Evolution / Schema Drift**: Handling changes in source schema without breaking downstream consumers. Supported by Delta Lake/Iceberg.
7. **Data Skew**: Uneven data distribution causing hotspots (e.g., in Spark joins or partitioning). Mitigation: salting, custom partitioning.
8. **Partitioning & Clustering**: Dividing data into manageable chunks (by date, region, etc.) for performance and pruning. Clustering for sort-order optimization (Snowflake/BigQuery).
9. **Medallion Architecture (Bronze/Silver/Gold)**: Layered lakehouse pattern — Bronze (raw), Silver (cleaned/conformed), Gold (aggregated/business-ready).
10. **Data Contracts**: Formal agreements on schema, SLAs, and quality between producers and consumers (helps with Data Mesh).
11. **Backpressure handling in streaming**.
12. **Quality gates and observability tools** (Great Expectations, Monte Carlo).
13. **Time Travel / Versioning + ACID in lakes** (Delta/Iceberg/Hudi).
14. **Zero-ETL / Reverse ETL**.
15. **Event-Driven Architecture** (vs batch/scheduled).

Interview Tip: These appear frequently in system design rounds. Always discuss failure modes, retries, and idempotency.

🏗️ 18. Architecture & Modern Patterns

1. **Lambda Architecture**: Batch + speed layer for handling both historical and real-time data (older but still discussed for trade-offs).
2. **Kappa Architecture**: All processing via streaming (simpler than Lambda, no separate batch layer).
3. **Modern Data Stack (MDS)**: Cloud-native tools emphasizing ELT, self-service (e.g., Fivetran + dbt + Snowflake + Looker). Shift from heavy ETL to transformation in warehouse. 2026 evolution includes dbt as standard, Airflow/Cosmos orchestration, and AI-assisted pipeline generation.
4. **dbt (data build tool)**: Modular SQL-based transformations with testing, documentation, and version control.

Interview Tip: Be ready to critique older patterns vs. lakehouse-first approaches.

⚙️ 19. Reliability & Observability (Heavily Tested)

1. **Data Observability**: Monitoring data for freshness, volume, schema changes, distribution anomalies (beyond basic pipeline monitoring). Pillars include freshness, volume, schema, distribution.
2. **Fault Tolerance & Retry Mechanisms**: Handling failures gracefully in distributed systems.
3. **Data Quality Gates / Checks**: Automated validation at different pipeline stages.
4. **Time Travel / Versioning**: Ability to query historical versions of data (supported in Delta Lake, Iceberg, Snowflake).
5. **ACID Transactions in Lakes**: How lakehouse formats (Delta, Iceberg, Hudi) bring ACID guarantees to object storage.

Interview Tip: As a manager, discuss how you build reliable platforms (SLAs, alerting, RCA processes).

📊 20. SQL & Modeling Depth (Frequently Tested)

1. **Window Functions**: RANK(), ROW_NUMBER(), LAG/LEAD, etc. — almost always appear in SQL rounds.
2. **Incremental Loading / Upsert / Merge**: Efficient ways to update data without full reloads (MERGE statement in modern warehouses).
3. **Sharding**: Horizontal partitioning across multiple databases/servers (vs partitioning within one table).
4. **Query Optimization**: CTEs, performance tuning on warehouses (partition pruning, clustering).

Interview Tip: Expect SQL screens even for managers. Practice real-world scenarios (running totals, deduplication, time-based analysis).

🧭 21. Leadership & Data Strategy (Critical for Manager Role)

1. **Data Engineering Vision & Roadmap**: Aligning technical roadmap with business/product goals.
2. **Team Scaling & Hiring**: Profiles, leveling, interview loops, onboarding.
3. **Stakeholder Management & Cross-Functional Collaboration**: With Data Science, Analytics, Product, Legal.
4. **Performance Management, Mentorship, Feedback**.
5. **Budgeting / FinOps / Resource Allocation**.
6. **OKRs & Success Metrics**: Reliability, adoption, cost efficiency, business impact.
7. **Change Management**: Large-scale migrations, adopting Data Mesh/lakehouse.
8. **Handling outages, influencing without authority, motivating teams**.

Interview Tip: Leverage your QA Manager experience (scaling processes, quality focus). Prepare STAR stories. Expect behavioral questions on leadership principles (especially Amazon-style).

🏛️ 22. System Design for Data Engineering (Make-or-Break for FAANG)

Common Prompts: Design a real-time analytics platform, Customer 360 pipeline at petabyte scale, or scalable ingestion/processing system.

Key Elements to Cover:

- Requirements gathering (volume, velocity, variety, latency, users).
- End-to-end architecture (ingestion → storage (Medallion layers) → transformation → serving).
- Trade-offs (batch vs. streaming, cost vs. freshness vs. accuracy).
- Scalability, reliability (idempotency, exactly-once, observability), security/governance.
- Failure modes, backfills, schema evolution, cost optimization.

Interview Tip: Practice verbalizing diagrams. Always clarify assumptions, discuss alternatives, and tie to business impact. Apply manager lens: How the design affects team processes and SLAs.

Study Recommendations

- **Prioritize**: Advanced Concepts (Section 17) + System Design (22) + Leadership & Strategy (21) first.
- **For every topic**: Prepare **Definition** → **Example** → **Trade-offs** → **When to use / not use** → **Manager implication** (scale, cost, team impact).
- **Practice**: 8–10 system design mocks, SQL scenarios (window functions, incremental loads), and behavioral STAR stories.
- **Hands-on**: Experiment with dbt + Snowflake/Databricks, CDC (Debezium + Kafka), and observability tools.

This version is now **clean, consistent, comprehensive, and ready to use** as your primary study notes. Everything from your input has been preserved and better organized.